



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Bhavana Kondakrindi
Roll Number	2520030082
Branch	CSE
Course Outcome	CO5 – Heap Structures & Stream Processing

Case Study Topic: Real-Time Stream Leaderboard Processing Using an Optimized Min-Heap
(Top-K Selection)

Work Done Summary:

In this case study, we implemented an optimized Min-Heap structure using a [PriorityQueue](#) to dynamically track and isolate the top- K most-listened-to artists from a continuous data stream. This technique minimizes memory consumption by avoiding full-dataset sorting, making it ideal for high-throughput streaming platforms.

Implementation Details:

1. Designed a structured Artist entity model to bind streaming metrics (monthly listener counts) deterministically.
2. Initialized a bounded Min-Heap variant configured to maintain exactly K top-tier elements dynamically.
3. Applied real-time stream evaluation logic: incoming data points are directly ignored if below the heap root minimum, or swapped in to eject the lower-bound threshold.
4. Extracted elements into a localized list and sorted descending to form a finalized scoreboard output.
5. Achieved an optimized algorithmic complexity bound of $O(N \log K)$ runtime and bounded $O(K)$ operational auxiliary storage space.

Result: Screenshots/output

```
# PS C:\Users\Bhavana> cd C:\Program Files\Java\jdk-11.0.10\bin
PowerShell> java -cp .\src\bin\*.jar -Djava.class.path=. -Djava.library.path=.
a:\AppData\Local\Temp\vscode\workspace\workspace\src\bin\*.jar
Processing 12 artists to find the top 5...

--- Final Top 5 Leaderboard ---
1. Artist10 (510)
2. Artist6 (480)
3. Artist8 (470)
4. Artist5 (470)
5. Artist3 (560)
PS C:\Users\Bhavana>
```

GitHub Link:

<https://github.com/bhavanakondakrindi/DSA-2-PROJECT/blob/main/CO5Assignment.java>